

Improving Patient Appointment Attendance and Healthcare Support Using Data (Initial Analysis Document)

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Project and Resource Review

This week's project is based on improving patient appointment attendance and healthcare support using data during the HealthConnect Clinic Experience Lab. As a data analyst, the HealthConnect_Appointment_Data and HealthConnect_Data_dictionary have been provided to guide my analysis. Through this data, I will be investigating the causes of patients missing scheduled appointments, factors enabling this behaviour, the inefficient use of appointment slots when patients fail to attend, the repetitive patient enquiries relating to appointments and clinic procedures, and the need to improve patient engagement and support. The main objective of this project is to determine how HealthConnect Clinic can use data to reduce missed appointments and improve patient support experience.

Since the HealthConnect Clinic Experience Lab is going to span multiple weeks, the main focus of this week is to understand, plan and some initial exploration. And so, as a data analyst, my main objective is to understand the appointment data and identify how it can be used to investigate the appointment attendance and no-show patterns.

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Data Quality Assessment

The HealthConnect_Appointment_Data dataset includes information relating to patient demographics, appointment details, booking information, previous appointment history, previous no-shows, reminder information, distance to the clinic, waiting time and appointment outcomes. The dataset consists of 18 columns and 5001 rows (with the first row being comprised of the column headers.).

The HealthConnect_Data_dictionary provides explanations of the variables contained in the HealthConnect_Appointment_Data dataset to further enhance the understanding of the said dataset. This dictionary consists of 5 columns and 19 rows (with the first row being comprised of the column headers.). The categories used to explain the variables include the data type, the description, an example and the notes for further elaboration. The variables in this case are the column headers in the aforementioned dataset.

The problems encountered in the dataset include:

- Inconsistent formatting for columns such as booking_date and the appointment_date where some dates are left aligned while others are right aligned.

- Wrong data types for columns such as the booking_date and the appointment_date where the data type was set to general.
- Blanks in the waiting time column and the distance to clinic column

On a brighter note, there are no duplicates in the appointment_id column (this is the column that uniquely identifies each appointment and so should not have any duplicates).

Relevant Business Questions

1. Which age group has the greatest no-show numbers?
2. Which reminder channels results in the most numerous no-shows?
3. What is the correlation between booking_lead_days and appointment attendance?
4. What is the relationship between the appointment day and the appointment attendance?
5. Does sending out reminders have an effect on the appointment outcome?
6. What is the relationship among the patient age, distance to the clinic and appointment outcome?
7. Does a greater number of previous appointment guarantee likeliness to attend one's appointment?
8. Which appointment time is most preferred by patients?
9. What is the effect of the waiting time on the appointment outcome?
10. Does the number of previous no-shows have an effect on the appointment outcome?

Potential KPIs

1. % no show rate (correlates with all the questions)
2. % attendance rate (correlates with all the questions)
3. % no show rate by reminder channel (correlates with question 2)
4. % no show rate by appointment day (correlates with question 4)
5. % no show rate by reminders sent (correlates with question 5)
6. % no show rate by age group (correlates with question 1)

Initial Analysis Approach

Since the data understanding and quality check has already been covered this week, I am going to have to start with the data cleaning in SQL to ensure that my data is ready to provide accurate insights into this patient attendance problem. The data cleaning is going to include handling missing data, fixing data types and creating key fields. The clean data will then undergo Exploratory Data Analysis to try to figure out the answers to my business questions and discover any possible limitations to my analysis.

Finally the data will be organised into a dashboard for visualisation on Tableau.

Factors that May Affect My Proposed Approach

- Cleaning the data might tamper with the accuracy of the data due to incomplete information.
- The data may not be enough to answer all the business questions.
- The patterns may be too irregular to determine a single distinct cause to the business problem.

- There may be quite a challenge in the analysis stage since SQL is very particular and specific on the instructions it receives. I am going to have to be very elaborate